

Carlos Manuel Martínez Vázquez

Computer Science Student @ Monterrey Institute of Technology

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SUMMARY

Computer Science student and incoming Quantitative Researcher with a strong foundation in statistics, optimization, and algorithmic modeling. Experienced in ML-based alpha signal extraction and robust cross-sector allocation using advanced econometrics.

EDUCATION

Monterrey Institute of Technology <i>B.S. in Computer Science and Technology (GPA: 3.7/4.0)</i> Honors: Academic Excellence Scholarship.	Querétaro, México Aug 2023 – June 2027
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EXPERIENCE

Quantitative Researcher Intern <i>Moreton Capital Partners</i> Incoming intern focusing on systematic commodity strategies and multi-horizon signal blending. Developing regime-aware portfolio optimization frameworks and risk-parity allocation models.	April 2026 – Present <i>Mexico City, México</i>
Research Software Developer <i>Instituto Via Diseño</i> Designed and implemented a relational database in PostgreSQL to model professors, classrooms, and student constraints for automated schedule generation and developed a scheduling algorithm based on constraint satisfaction and backtracking heuristics to minimize class conflicts and optimize time-slot allocation. Integrated algorithmic results into a full-stack visualization platform using Node.js, React, and TypeScript, enabling real-time edits and analytics for 150+ users.	Feb 2025 – May 2025 <i>Querétaro, México</i>

RESEARCH & PROJECTS

Hybrid Asset Pricing: LLM Sentiment & Bi-LSTM <i>Working Paper</i> Authoring a research paper comparing lexicon-based models (VADER) vs. LLM-driven sentiment analysis in high-volatility markets. Developing a hybrid architecture utilizing a Multilayer Perceptron (MLP) to filter stochastic noise and isolate high-fidelity alpha signals.	Dec 2025 – Present
Hierarchical Commodity Portfolio Construction Developed a two-stage backtesting engine for 10 commodity sectors over 162 weekly periods with SLSQP min variance optimization algorithm that achieved max 7.7% annual volatility. Enforced KKT optimality to maintain stable weights and model transactions and achieved a 4.02 Adjusted Sharpe (Andrew W Lo 2002) and 2.1 Calmar Ratio by implementing Ledoit-Wolf shrinkage to regularize covariance matrices.	Mar 2026
AquaHub: Predictive Drought Modeling & Crowdsourcing Engineered a predictive ML model for drought detection in CDMX, processing 960+ training samples to identify water scarcity hotspots and validated a 5-day sliding window architecture, achieving 70–85% accuracy in preliminary drought event detection. Designed a validation pipeline to cross-reference crowdsourced reports against meteorological norms, covering a demographic of 17.7M residents.	Feb 2026
Bloomly: Global Bloom Detection System Engineered a LightGBM-based predictive model leveraging multi-spectral satellite and meteorological data and conducted feature selection and dimensionality reduction across 44 ecological indicators to improve classification precision and evaluated model performance via AUC and F1 metrics under temporal cross-validation for predictive stability. Deployed Flask APIs with interactive 3D data visualization (Three.js) for spatial analysis of global bloom patterns.	Oct 2025

TECHNICAL SKILLS

Languages & Quant: Python, C++, R, SQL, JavaScript, scikit-learn, LightGBM, pandas, NumPy, Mesa
Dev & Cloud: React, Next.js, Node.js, Flask, PostgreSQL, AWS, Git **Spoken:** Spanish (Native), English (C1)

RESEARCH INTERESTS

Quantitative modeling, NLP for finance, time-series forecasting, portfolio optimization, and multi-agent simulation of market dynamics.